



Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E Computer Science and Engineering

Course Code & Title: CS8792 Cryptography and Network Security

Name of the Faculty member (s): Dr.R.Venkatesh, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic:** Unit I / Steganography
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO3
- **Activity Chosen:** Worksheet
- **Justification:**

Quite apart from transferring theoretical concepts to students, it is great practice to have a prototype, presentation, or real-world examples. Worksheets are a type of model in which the concept of Steganography can be easily transferred and students can grasp the concept with a real understanding.

Time Allotted for the Activity:

- **Procedure of the Implementation:**

A random paragraph was chosen and printed, with some characters punctured with a safety pin. When the punctured characters are read in order, a secret message is produced. Students were instructed to find the hidden message in the given paragraph.

- **Details of the Implementation**

- First of all, teacher taught the topics and provide the worksheet with the pin punctures, without telling any information about the pin puncture to the students.
- Initially, students were not able to find the pin punctures, but some of the students revised the steganography content, and they identified the pin punctures while showing in front of the light.
- Finally, the students shared their ideas and identified confidential messages bases on their worksheet.
- The Figure 1 shows the Worksheet activity done by the students.
- Finally, teacher and students discussed the answers of all questions at the end of the activity.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO4	PO11	PSO2
CO1	3	2	3	1	2

(1 – Low 2 – Moderate 3 – High)

• **Images / Screenshot of the practice:**



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Work Sheet I

The Challenges of Computer Security

Computer and network security is both fascinating and complex. Some of the reasons follow:

1. Security is not as simple as it might first appear to the novice. The requirements seem to be straightforward; indeed, most of the major requirements for security services can be given self-explanatory, one-word labels: confidentiality, authentication, nonrepudiation, or integrity. But the mechanisms used to meet those requirements can be quite complex, and understanding them may involve rather subtle reasoning.
2. In developing a particular security mechanism or algorithm, one must always consider potential attacks on those security features. In many cases, successful attacks are designed by looking at the problem in a completely different way, therefore exploiting an unexpected weakness in the mechanism.
3. Because of point 2, the procedures used to provide particular services are often counterintuitive. Typically, a security mechanism is complex, and it is not obvious from the statement of a particular requirement that such elaborate measures are needed. It is only when the various aspects of the threat are considered that elaborate security mechanisms make sense.
4. Having designed various security mechanisms, it is necessary to decide where to use them. This is true both in terms of physical placement (e.g., at what points in a network are certain security mechanisms needed) and in a logical sense (e.g., at what layer or layers of an architecture such as TCP/IP [Transmission Control Protocol/Internet Protocol] should mechanisms be placed).
5. Security mechanisms typically involve more than a particular algorithm or protocol. They also require that participants be in possession of some secret information (e.g., an encryption key), which raises questions about the creation, distribution, and protection of that secret information. There also may be a reliance on communications protocols whose behaviour may complicate the task of developing the security mechanism. For example, if the proper functioning of the security mechanism requires setting time limits on the transit time of a message from sender to receiver, then any protocol or network that introduces variable, unpredictable delays may render such time limits meaningless.
6. Computer and network security is essentially a battle of wits between a perpetrator who tries to find holes and the designer or administrator who tries to close them. The great advantage that the attacker has is that he or she need only find a single weakness, while the designer must find and eliminate all weaknesses to achieve perfect security.
7. There is a natural tendency on the part of users and system managers to perceive little benefit from security investment until a security failure occurs.

Decrypted Confidential Message: Please help me & I'm wounded.



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Work Sheet II

the relative frequencies of individual letters exhibit a much greater range than that of digrams, making frequency analysis much more difficult. For these reasons, the Playfair cipher was for a long time considered unbreakable. It was used as the standard field system by the British Army in World War I and still enjoyed considerable use by the U.S. Army and other Allied forces during World War II. Despite this level of confidence in its security, the Playfair cipher is relatively easy to break, because it still leaves much of the structure of the plaintext language intact. A few hundred letters of ciphertext are generally sufficient. One way of revealing the effectiveness of the Playfair and other ciphers is shown in Figure 3.6. The line labeled *plaintext* plots a typical frequency distribution of the 26 alphabetic characters (no distinction between upper and lower case) in ordinary text. This is also the frequency distribution of any monoalphabetic substitution cipher, because the frequency values for individual letters are the same, just with different letters substituted for the original letters. The plot is developed in the following way: The number of occurrences of each letter in the text is counted and divided by the number of occurrences of the most frequently used letter. Using the results of Figure 3.5, we see that e is the most frequently used letter. As a result, e has a relative frequency of 1, of 9.056/12.702 $\approx$ 0.72, and so on. The points on the horizontal axis correspond to the letters in order of decreasing frequency. Figure 3.6 also shows the frequency distribution that results when the text is encrypted using the Playfair cipher. To normalize the plot, the number of occurrences of each letter in the ciphertext was again divided by the number of occurrences of e in the plaintext. The resulting plot therefore shows the extent to which the frequency distribution of letters, which makes it trivial to solve substitution. An Army Signal Corp officer, Joseph Mauborgne, proposed an improvement to the Vernam cipher that yields the ultimate in security. Mauborgne suggested using a random key that is as long as the message, so that the key need not be repeated. In addition, the key is to be used to encrypt and decrypt a single message, and then is discarded. Each new message requires a new key of the same length as the new message. Such a scheme, known as a one-time pad, is unbreakable. It produces random output that bears no statistical relationship to the plaintext.

Decrypted Confidential Message: Meet capt at lobby.



Figure 1 – Students trying to find out the Hidden Message

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO4	PO11	PSO2
	3	2	3	1	2
Justification for correlation	Basic Computer Science Engineering and mathematics Knowledge for performing classical encryption is highly correlated	Analyzing various classical encryption algorithms is moderately correlated	Investigate various aspects of cryptanalysis is highly correlated	Deploying classical encryption security principles in the projects are weakly correlated	Providing solution to real world problems with classical encryption principles are moderately correlated

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

Students believe that their self-learning has improved. They realized how to communicate with the concept of secret writing.

❖ **Benefit of the Practice**

During the activity, student's coordination and attention were highly increased and same has been appreciated. They learnt how ancient cryptology methods were practiced over the period decades.

❖ **Challenges faced in implementation:**

The activity should be carried out with dedicated hour. Since the activity clubbed with other topics, class room arrangement for proper lighting consumes little time.

References:

- https://www.teachingenglish.org.uk/sites/teacheng/files/Innovation_is_Great_student_worksheets.pdf
- William Stallings, Cryptography and Network Security: Principles and Practice, PHI 3rd Edition, 2006.

Signature of Faculty Member**HOD-CSE**